Task 5: Decision Trees and Random Forests

Important outputs from the executed Google Colab notebook

Classification dataset: Wine Recognition (178 samples, 13 features, 3 classes)

Regression dataset: Diabetes (442 samples, 10 features)

Objective: Train tree-based classification and regression models, study overfitting, compare models, analyze feature importance, and evaluate classification models using cross-validation.

The following pages contain the most relevant results and visualizations generated by the notebook.

1. Wine Dataset Preview and Class Distribution

```
Dataset Shape: (178, 14)
Number of Features: 13
Target Classes:
class_0 : 59
class_1 : 71
class_2 : 48

Dataset Preview (first 5 records):

   alcohol  malic_acid   ash  ...  od280/od315_of_diluted_wines  proline  target
0    14.23        1.71  2.43  ...                          3.92   1065.0       0
1    13.20        1.78  2.14  ...                          3.40   1050.0       0
2    13.16        2.36  2.67  ...                          3.17   1185.0       0
3    14.37        1.95  2.50  ...                          3.45   1480.0       0
4    13.24        2.59  2.87  ...                          2.93    735.0       0

[5 rows x 14 columns]
```

2. Training and Testing Split

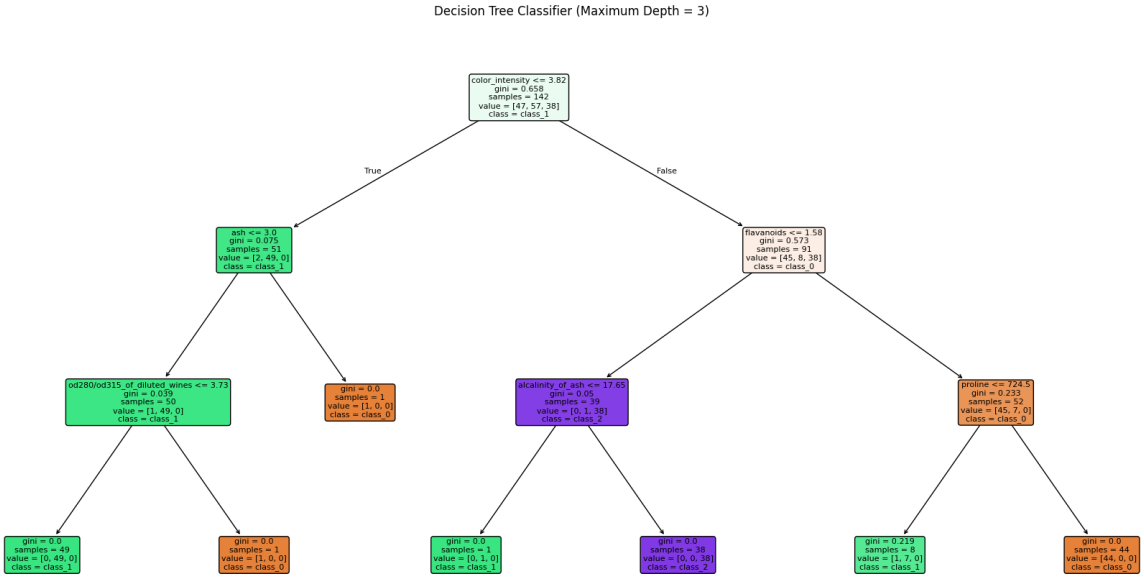
```
Training Data: (142, 13)
Testing Data: (36, 13)
Training Samples: 142
Testing Samples: 36
```

3. Decision Tree Overfitting Comparison

```
Training vs Testing Accuracy:

                          Model  Training Accuracy  Testing Accuracy
0  Decision Tree (unrestricted)              1.000            0.9444
1   Decision Tree (max_depth=3)              0.993            0.9444
```

4. Decision Tree Visualization



5. Decision Tree and Random Forest Comparison

```
Decision Tree vs Random Forest:

                        Model  Training Accuracy  Testing Accuracy
0  Decision Tree (max_depth=3)              0.993            0.9444
1    Random Forest (200 trees)              1.000            1.0000
```

6. Random Forest Confusion Matrix and Classification Report

```
Confusion Matrix:
[[12  0  0]
 [ 0 14  0]
 [ 0  0 10]]

Classification Report:
              precision    recall  f1-score   support

     class_0       1.00      1.00      1.00        12
     class_1       1.00      1.00      1.00        14
     class_2       1.00      1.00      1.00        10

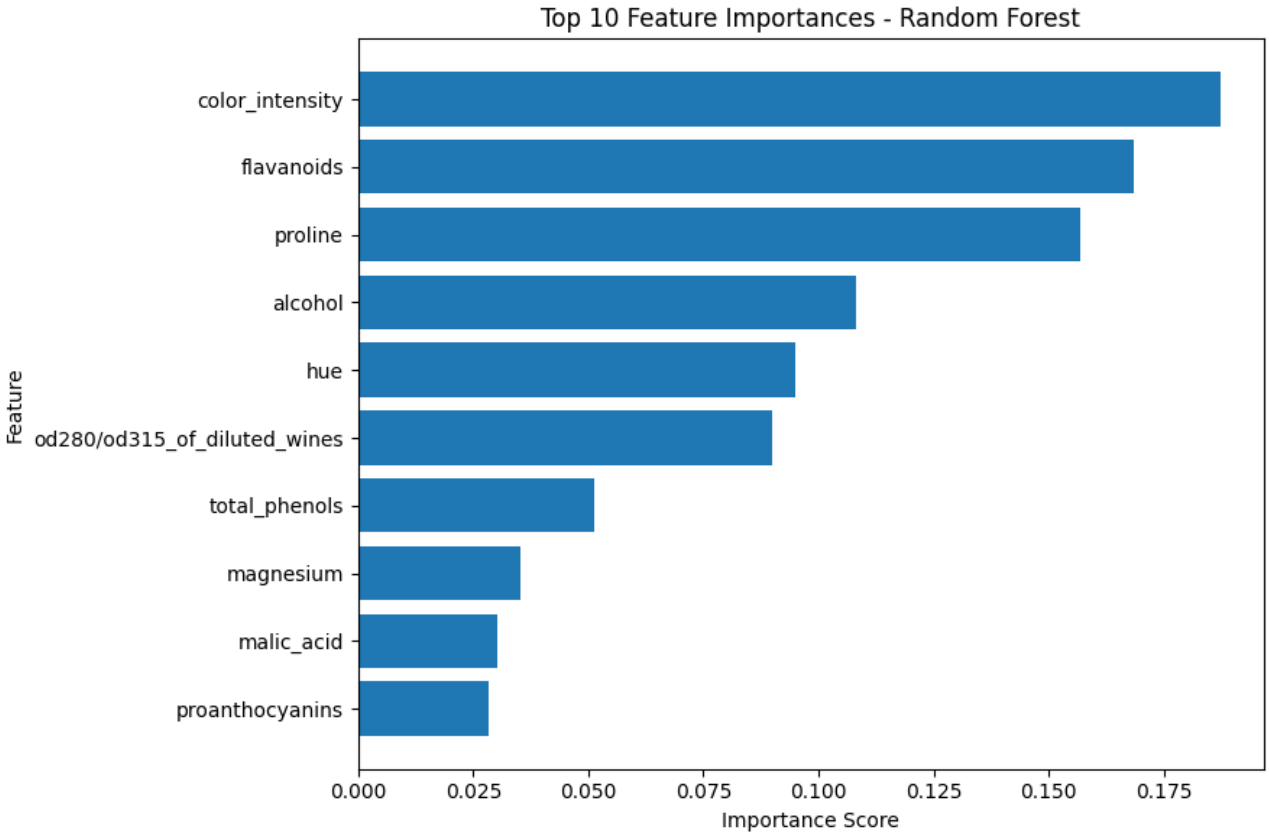
    accuracy                           1.00        36
   macro avg       1.00      1.00      1.00        36
weighted avg       1.00      1.00      1.00        36
```

7. Top 10 Feature Importance Values

```
Top 10 Important Features:

                         Feature  Importance
9                color_intensity      0.1874
6                     flavanoids      0.1684
12                       proline      0.1570
0                        alcohol      0.1082
10                           hue      0.0950
11  od280/od315_of_diluted_wines      0.0899
5                  total_phenols      0.0513
4                      magnesium      0.0352
1                      malic_acid      0.0302
8                proanthocyanins      0.0283
<Figure size 900x600 with 1 Axes>
```

8. Feature Importance Graph



9. 5-Fold Cross-Validation Results

```
5-Fold Cross-Validation Results:
                     Model  Fold 1  Fold 2  Fold 3  Fold 4  Fold 5  Mean Accuracy
Decision Tree (max_depth=3)  0.7931  0.8966  0.9286  0.8929  0.9643         0.8951
  Random Forest (200 trees)  0.9310  1.0000  1.0000  1.0000  0.9643         0.9791

Decision Tree Mean CV Accuracy: 0.8951
Random Forest Mean CV Accuracy: 0.9791
```

10. Diabetes Dataset Preview

```
Regression Dataset Shape: (442, 11)
Number of Regression Features: 10

Regression Dataset Preview:
                0          1           2           3           4
age      0.038076  -0.001882    0.085299   -0.089063    0.005383
sex      0.050680  -0.044642    0.050680   -0.044642   -0.044642
bmi      0.061696  -0.051474    0.044451   -0.011595   -0.036385
bp       0.021872  -0.026328   -0.005670   -0.036656    0.021872
s1      -0.044223  -0.008449   -0.045599    0.012191    0.003935
s2      -0.034821  -0.019163   -0.034194    0.024991    0.015596
s3      -0.043401   0.074412   -0.032356   -0.036038    0.008142
s4      -0.002592  -0.039493   -0.002592    0.034309   -0.002592
s5       0.019907  -0.068332    0.002861    0.022688   -0.031988
s6      -0.017646  -0.092204   -0.025930   -0.009362   -0.046641
target  151.000000  75.000000  141.000000  206.000000  135.000000
```

11. Regression Training and Testing Split

```
Regression training data: (353, 10)
Regression testing data: (89, 10)
Both regression models trained successfully!
```

12. Regression Model Comparison (MAE, RMSE and R^2)

```
Regression Model Comparison:
                              Model     MAE    RMSE  R2 Score
Decision Tree Regressor (max_depth=4) 47.0454 59.7408    0.3264
  Random Forest Regressor (200 trees) 44.2761 54.4612    0.4402
```

13. Random Forest Regressor: Actual vs Predicted

